MunimJi — Features & Build Checklist

Nov 4, 2026 · @Harsh Gupta

Overview

MunimJi lets shop owners send bill photos or PDFs into a Telegram group, where AI agents read and GST-check them, and a CA approves everything before filing.

Item	Detail
Hackathon	Flip the Switch (WeMakeDevs x DevTools Academy x Switch)
Track	Track 1: Best Business Workflow
Deadline	Nov 7, 2026, 11:59 PM IST (aim to submit by evening)
Core rule	AI reads, code checks, human decides
Data	All bills, shops and GST numbers are synthetic

Features

Build the must-haves first; they alone make a complete, judgeable demo.

Must-have

Feature	What it does	Who uses it	Where
Private shop groups	One Telegram group per shop; shops never see each other's bills	Shop owner	Shop group
Bill photo / PDF intake	Owner drops a bill photo or PDF (incl. multi-page or e-invoice PDFs); agent acknowledges within seconds	Shop owner	Shop group

Feature	What it does	Who uses it	Where
Bill Reader agent	Reads text PDFs directly and photos or scanned PDFs as images; extracts supplier, GSTIN, date, bill number, items, amount, tax	AI	Shop group
Confirm or correct	Owner approves the extracted details or types a fix	Shop owner	Shop group
GST checks (code)	GSTIN present and valid format + check digit; CGST/SGST vs IGST by state code; totals add up; duplicate bill; missing fields	System	Backend
Issue routing	Simple problems go back to the owner; serious ones go to the CA	AI + system	Both rooms
Follow-up agent	Polite reminder to the owner with the rupee amount at risk	AI	Shop group
CA office room	One place to see problems across all shops and approve or reject	CA	CA room
Approve / reject / ask supplier	CA decides on serious issues with one reply	CA	CA room
Month-end summary	Return Preparer totals clean bills, open issues and tax credit	AI	CA room
Summary file export	Excel or JSON file the CA reviews and uploads herself	CA	CA room
One-command setup	Template + seed data so judges can reproduce the demo	Judges	Repo

Nice-to-have

Feature	What it does	Who uses it
Hindi / Hinglish replies	Agents reply in the language the owner writes in	Shop owner

Feature	What it does	Who uses it
Blurry photo detection	Agent asks for a clearer photo instead of guessing	Shop owner
Deadline reminders	"5 days left; last month you sent ~70 bills, you've sent 64"	Shop owner
Per-shop status command	Owner types a command to see bills sent, clean and pending	Shop owner
CA dashboard command	One command lists every shop's status for the month	CA

Stretch (only if time is left)

Feature	What it does
Voice notes	Owner speaks in Hindi; agent transcribes and links it to the bill
Supplier chase	Agent drafts a message to the supplier for a corrected bill, sent only after owner approval
Trust levels	Agent earns the right to auto-accept clean bills from repeat suppliers after enough CA approvals
Second shop in demo	Shows privacy between shops and the CA handling many clients

Build checklist

One day per layer: setup today, bill pipeline tomorrow, routing and summary on Nov 6, polish and submit on Nov 7.

Nov 4 (today): setup

- [ ] Register for the hackathon on WeMakeDevs
- [ ] Install Docker and Switch Console; check your OS build works (Windows/Linux are early access)

- [ ] Run the 10-minute quickstart; get Claude Code replying in a Mattermost room
- [ ] Connect one Telegram group (test early; fall back to Mattermost if it fails)
- [ ] Connect Claude Code to the Switch docs MCP and read the agent protocol
- [ ] Confirm an agent only sees the rooms it is added to
- [ ] Create the repo with a README skeleton
- [ ] Make 15-20 synthetic bills as photos and PDFs, including one multi-page PDF (clean, missing GSTIN, wrong tax type, wrong total, duplicate, blurry)
- [ ] Generate fake GSTINs with a correct format and check digit
- [ ] Optional: claim a Track 3 issue labelled hackathon

Nov 5: bill pipeline

- [ ] Postgres tables: shops, bills, issues
- [ ] FastAPI service the agents call to save and check bills
- [ ] Bill Reader agent: photo or PDF in (extract text from text PDFs, convert scanned PDF pages to images), structured details out, posted back for confirmation
- [ ] Owner confirm / correct flow
- [ ] GST checks in code: GSTIN present, format, check digit
- [ ] GST checks in code: CGST+SGST vs IGST from state codes
- [ ] GST checks in code: totals, required fields, duplicate detection
- [ ] Unit tests for every check against the synthetic bills

Nov 6: routing, CA room, summary

- [ ] Create the CA office room with the CA, GST Checker and Return Preparer
- [ ] Route simple issues to the shop group via the Follow-up agent
- [ ] Route serious issues to the CA room with approve / reject / ask supplier
- [ ] Record the CA's decision on the bill and close the issue
- [ ] Return Preparer month-end summary message
- [ ] Export the summary as Excel or JSON
- [ ] Hindi / Hinglish replies (nice-to-have)
- [ ] Switch template YAML that creates all agents and rooms
- [ ] Seed script that loads shops and synthetic bills
- [ ] Full end-to-end dry run of the demo script

Nov 7: polish and submit

- [ ] Fix bugs found in the dry run
- [ ] Fresh-machine test: clone, one command, demo works
- [ ] Record the 3-minute demo video
- [ ] Finish the README (see submission checklist)
- [ ] Submit by evening, well before 11:59 PM IST
- [ ] Optional: blog post, #HowISwitch post, Wispr scorecard

Submission checklist

The repo, README and video are what judges see; each judging criterion should be answerable from them in under a minute.

What to submit

- [ ] Public GitHub repo
- [ ] README: problem, demo GIF, architecture diagram (shop groups, Switch, agents, CA room)
- [ ] README: a section titled "How Switch is used" naming the rooms, agents and the protocol
- [ ] README: one-command setup and the seed data
- [ ] README: "All data is synthetic" and "AI prepares, CA files"
- [ ] Demo video, about 3 minutes
- [ ] Submission form filled on WeMakeDevs

Judging self-check

Criterion	Question judges ask	How MunimJi answers it
Use of Switch	Would it stop working without Switch?	Private Telegram rooms per shop, a shared CA room, agents carrying issues between them
Technical execution	Can someone clone and run it?	Template YAML + seed script + tested rule-based checks

Criterion	Question judges ask	How MunimJi answers it
Creativity and impact	Fresh idea, real problem?	Monthly GST bill chaos for Indian small shops and their CAs
Presentation	Do the video and README show Switch clearly?	Video follows one bill from photo to CA approval; README diagram

Demo video beats

1. The problem: a bill pile and a CA chasing shop owners
2. Owner sends a bill photo, then a PDF invoice; agent reads both; owner confirms
3. Bill with no GSTIN; Follow-up agent flags the rupee amount at risk
4. Wrong tax type goes to the CA room; CA chooses "ask supplier"
5. Duplicate bill caught
6. Month-end summary; CA approves and downloads the file
7. Architecture slide and "AI prepares, CA decides"

Cut list if behind schedule

If you are behind on the evening of Nov 6, cut in this order and keep the core loop: photo, checks, CA approval, summary.

1. Voice notes and supplier chase (stretch)
2. Deadline reminders and status commands
3. Hindi replies (mention as planned in the README)
4. Telegram: run the whole demo on Mattermost and note Telegram as supported
5. Excel export: show the summary message and export JSON only

Never cut: the GST checks in code, the CA approval step, one-command setup, or the demo video.